

SLSTP FINAL PRESENTATION

National Aeronautics and
Space Administration



Synthetic transcriptomics for mouse spaceflight

Can generated RNA-seq improve tissue-specific FLT vs GC analysis?

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Biological & Physical Sciences



QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

1,610

biological profiles

75

accessions

835 FLT

775 GC

ARCHS4 mouse

997,515

profiles audited

17,244

selected

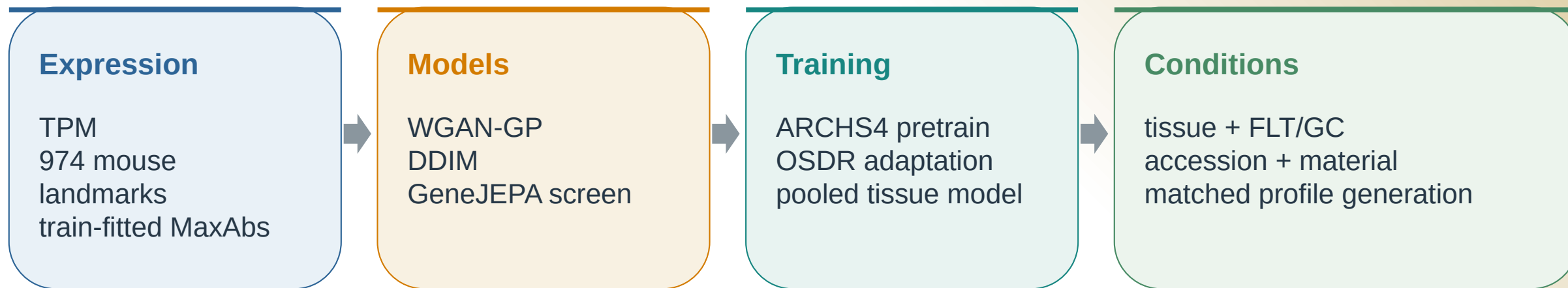
20 tissues

- Mission, strain, material and assay differences can resemble a flight effect.
- The generator must preserve tissue and FLT/GC structure without memorizing samples.
- Statistical evidence must still come from observed OSDR profiles.

METHOD

We compared three model families and used conditional diffusion

The model architecture followed each paper; preprocessing, harmonization and conditioning remained configurable.



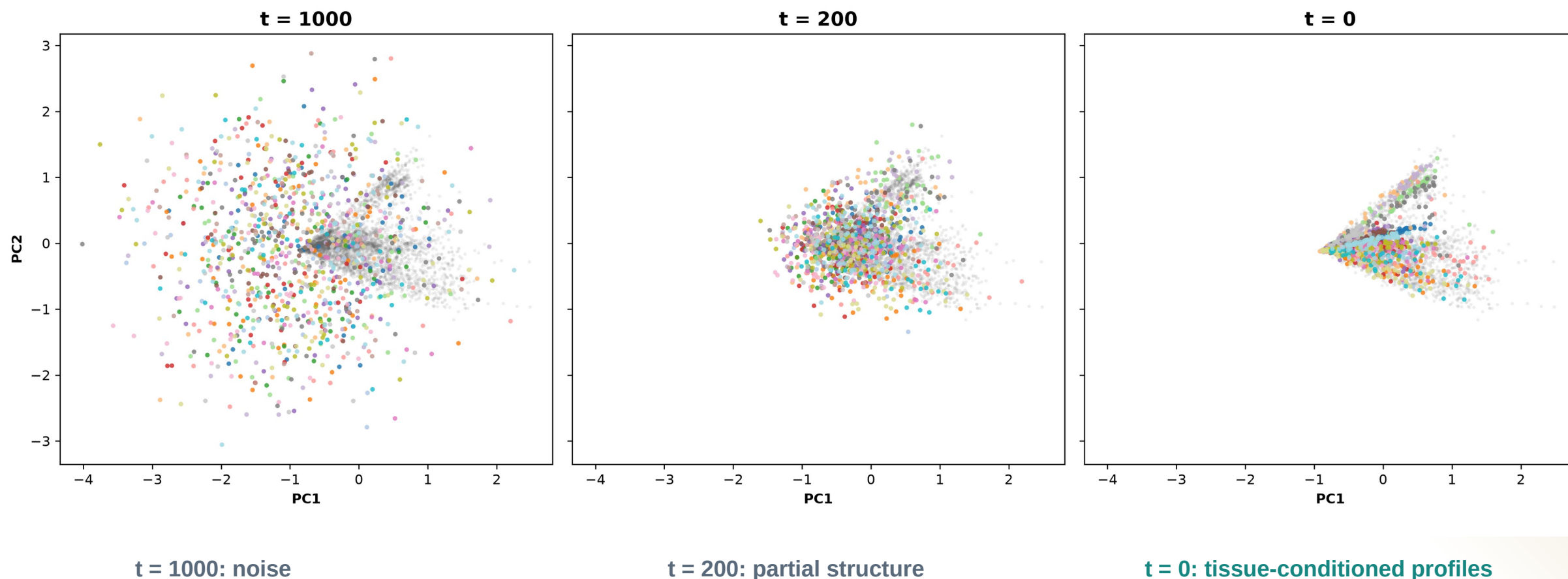
Selected branch **No global batch correction.** Accession was supplied as a model condition so study structure was represented instead of erased.

Nine liver harmonization arms and a 463-row experiment plan were screened before full training.

DIFFUSION

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



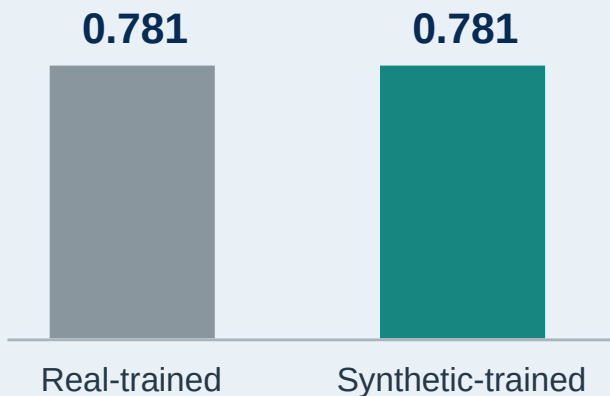
VALIDATION

DDIM matched expression and was harder to distinguish from real data

AA closer to 0.5 means a classifier has less success separating real and generated profiles; lower FD is better.

ARCHS4 tissue probe

Balanced accuracy on held-out real profiles



Metric

WGAN-GP

DDIM

Reading

Correlation

0.976

0.974

similar

F1

0.985

0.997

higher

Adversarial accuracy

0.636

0.475

closer to 0.5

FD / real P95

0.144

0.074

lower

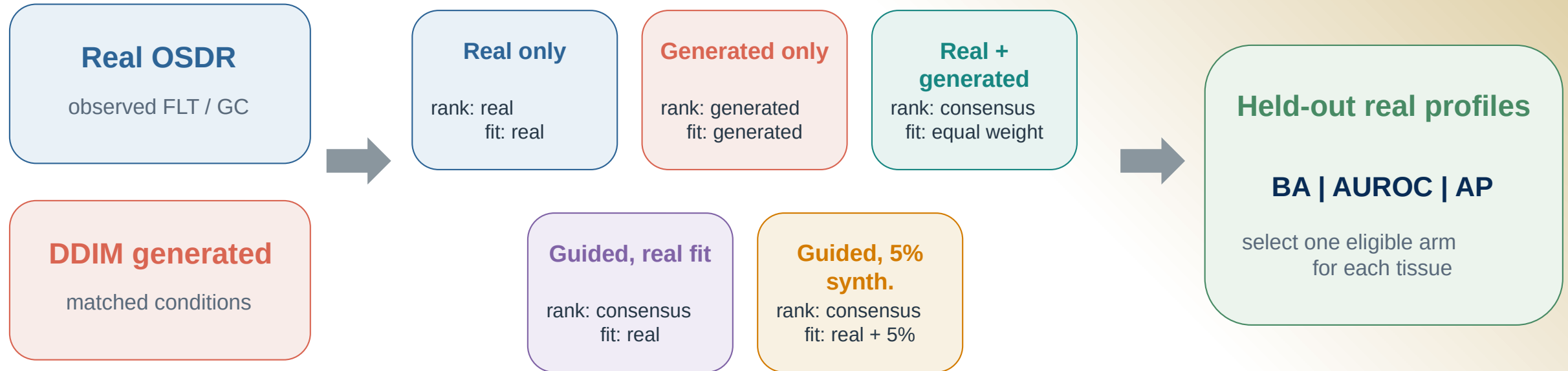
Use DDIM downstream

Lower separability and distributional distance, with comparable correlation and higher F1.

ANALYSIS

Synthetic profiles entered the analysis in five different ways

Every arm was judged on held-out real profiles before synthetic attribution was allowed.



Biology check FLT vs GC effects, random-effects meta-analysis and BH FDR use real OSD R profiles only.

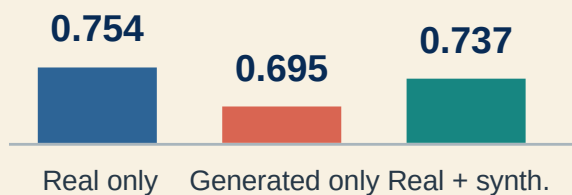
Synthetic profiles never increase animal n.

Pooling tissues hid useful signal

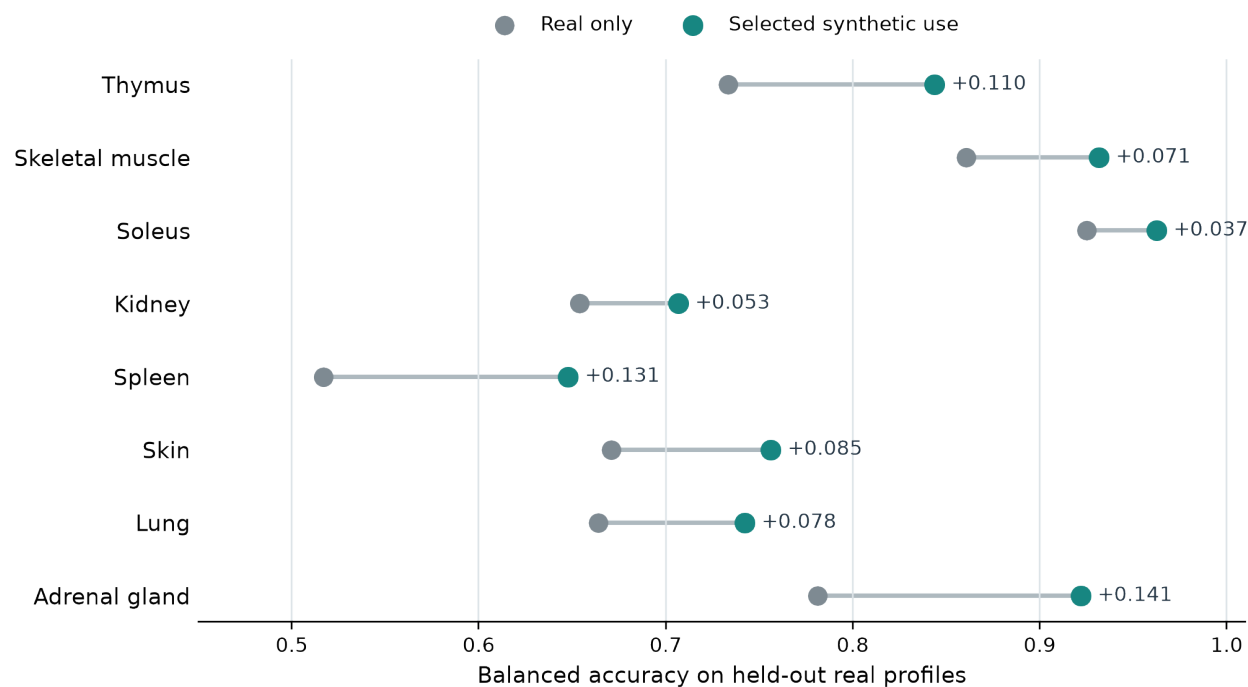
A single FLT/GC classifier did not improve, but tissue-specific synthetic use often did.

Pooled across tissues

Balanced accuracy



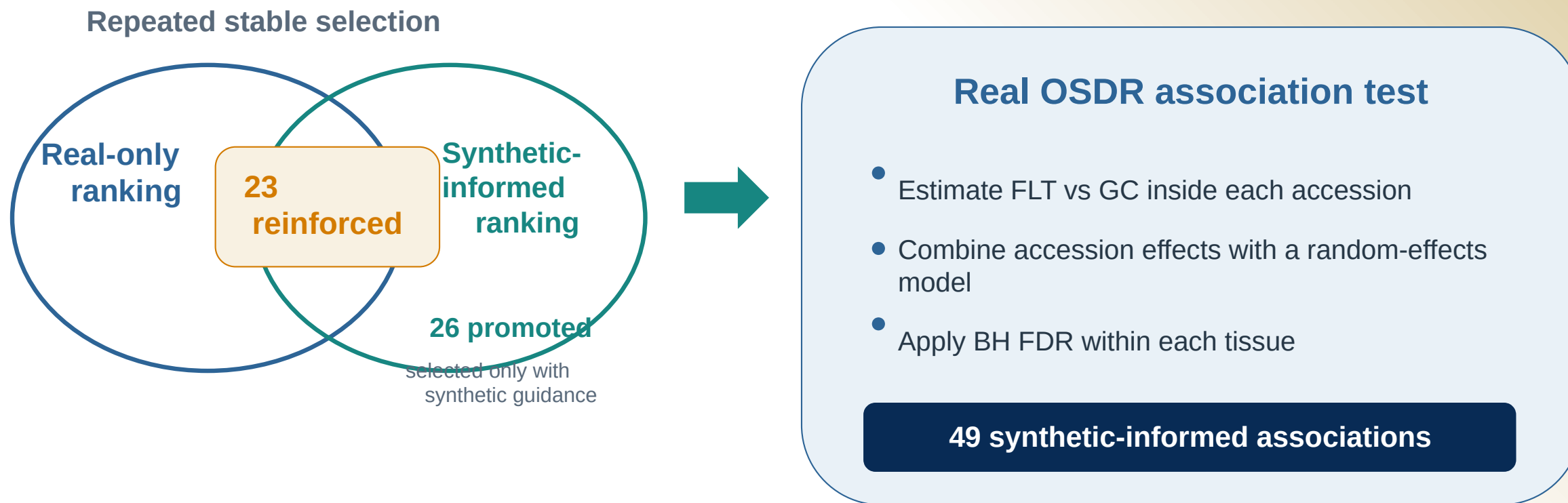
Selected use within each tissue



The useful arm differed by tissue.

Synthetic guidance changed ranking, not statistical evidence

Promoted and reinforced describe repeated feature selection; both require a supporting effect in real OSD data.



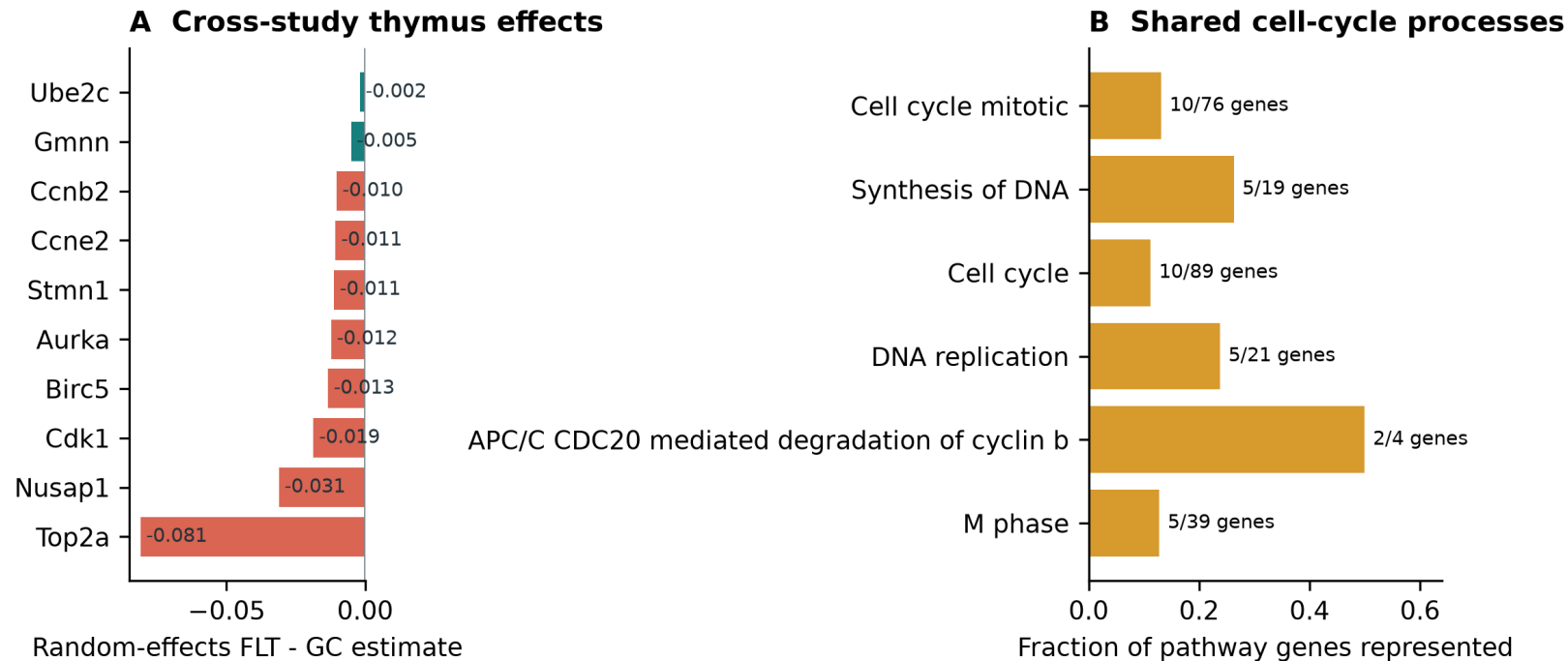
These are 49 tissue-gene associations among 459 BH-FDR rows. A gene can appear in more than one tissue.

THYMUS

Thymus points to lower proliferative renewal

Synthetic guidance organized real-data associations into a coherent mitotic and DNA-replication panel.

Synthetic-informed thymus genes converge on a flight-lower mitotic program



16 synthetic-informed
BH-FDR associations

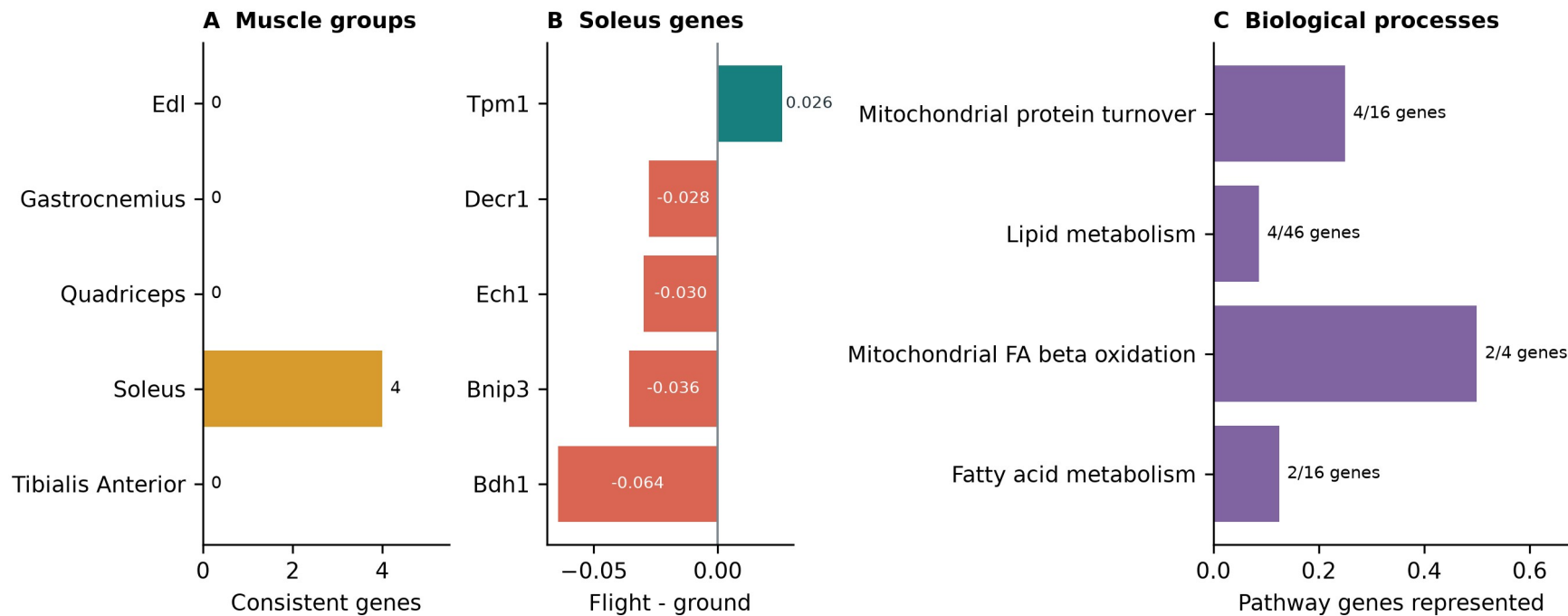
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Bulk RNA-seq cannot separate cell loss from lower transcription

Soleus reinforces a mitochondrial and lipid program

This result strengthens an existing real-data pattern rather than promoting a new soleus gene.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 -> 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

OTHER TISSUES

Kidney adds a focused signal; other tissues remain exploratory

The strongest process-level stories were thymus and soleus, but several tissues produced narrower candidates.

Kidney	Inpp4b promoted, Slc37a4 reinforced; both higher in flight	Focused secondary
Spleen	Rai14, Ptprk and Myl9 promoted; Loxl1 reinforced; no Reactome enrichment	Exploratory
Skin + adrenal	Plscr1 higher in skin; Psmb8 and Tspan4 lower in adrenal	Exploratory
Lung + retina	Predictive gains without a synthetic-informed BH-FDR gene	Prediction only
Liver + EDL + quadriceps	Real-only arm retained	No synthetic claim

Independent missions are still needed before any candidate becomes a transferable spaceflight biomarker.

TAKEAWAYS

Synthetic data helped most as a tissue-specific prior

The generator created useful structure, not new biological replication.

1 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

2 Use

Pooled augmentation failed.
Tissue-specific ranking and low-weight training were more useful.

3 Interpret

Thymus and soleus formed the clearest programs; kidney supplied a focused secondary pair.

Next test

Hold out complete accessions, then validate the thymus panel, soleus metabolism and kidney *Inpp4b* / *Slc37a4* in independent data.

Generated profiles never enter the biological sample count or the BH-FDR test.

Thank you

Questions?

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Mentor

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Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.